

Approx-PSI: Efficient Differentially Private Set Intersection-based Analysis

Jian Long, Ziyuan Liang, Chao Li, Fan Zhang, *Member, IEEE*, and Zhan Qin, *Member, IEEE*

Abstract—The Private Set Intersection (PSI) functionality is valuable which allows different organizations to collaborate in privacy-sensitive environments, and its intersection is highly valuable in analytics. With similar business interests, they can securely compute intersection and its associated values for insightful trends and patterns, which can be considered as intersection-based analysis. However, in such settings, existing PSI schemes are too costly or inadvertently leak sensitive information. Our proposed scheme, Approx-PSI, employs Oblivious Pseudo-Random Functions (OPRF) to provide two-party efficient PSI in a semi-honest model and introduces “noise” into the intersection through differential privacy (DP) to ensure individual privacy. Moreover, the noised intersection for approximate analysis has more flexibility and adaptability in both associated values and desired functions. Compared to existing schemes, Approx-PSI achieves at least $1.5 - 3\times$ faster than the most efficient prior protocols, and up to $77 - 81\times$ faster in certain settings, while reducing communication by at least 90%. Thus, the above benefits make our scheme more practical in the real-world.

Index Terms—Private Set Intersection, Randomized Response, Differential Privacy

I. INTRODUCTION

DATA sharing and collaborative analysis among different organizations have gained significant attention for enabling information exchange and unlocking the potential of data. Private Set Intersection (PSI) [1]–[5] is a fundamental cryptographic protocol that allows two parties to jointly compute the intersection of their datasets without revealing any elements that do not belong to the intersection. Examples include private contact tracing [6], where two parties identify shared time-location records indicating potential disease exposure, and private mobile contact discovery [7], [8], which finds overlapping contacts from users’ address books or logs.

However, most standard PSI protocols are designed to securely learn the intersection elements. This is insufficient for many practical applications, such as securely measuring ad conversions in advertising, analyzing shared patient metrics across hospitals, or aggregating behavioral statistics over overlapping users, where further analysis over the intersection is essential. Besides, these protocols raise privacy concerns due to the inherent disclosure of the intersection elements.

For example, if one party is a medical institution and the other is an insurance company, knowing that a specific person is in both datasets could reveal sensitive health information. This issue becomes more pronounced when the intersection involves highly specific data, like rare diseases where even a small overlap could disclose critical personal details.

To meet the needs, various schemes extend the basic functionality by enabling further analysis on the intersection and its *associated values* (also called *payload*), while ensuring that the intersection remains undisclosed. Notably, the associated values refer to additional data linked to each element, such as user attributes, transaction amounts, or behavioral metrics. These variants allow two parties, which hold the dataset of X and Y , to compute a specific function F over the intersection and its values, i.e. $F(X \cap Y)$.

In more detail, *Circuit-based PSI (C-PSI)* [4], [5] serves as a general-purpose framework for executing arbitrary functions on the intersection. Its flexibility makes it suitable for scenarios requiring complex analytics. Meanwhile, *PSI-cardinality (PSI-CA)* [22], a variant of PSI, is an aggregate functionality that solely focuses on computing the cardinality (size) of the intersection.

Private Intersection-Sum-with-Cardinality (PIS) [17] demonstrates how summing associated values within an intersection can safely aggregate metrics like revenue in online ads. Similarly, *PSI-stats* [20] enables computation of the mean or variance of the values. *Private Computing on Set Intersection (PCSI)* [21] proposes a protocol to securely perform cardinality and sum in different settings.

These approaches demonstrate a wide range of scenarios for analyzing an intersection and its values, which can be considered as *intersection-based analysis*. Such analysis can include cardinality, sum, mean, and variance, or even more complex functions on its values.

Despite their capability, these approaches above (Table I), based on garbled circuits, public-key encryption, and homomorphic encryption, require encrypting the values and exchanging encrypted data. They consume high computation and communication costs, especially for large datasets. Additionally, there is a risk that only disclosing cardinality or sum is also vulnerable to privacy attacks [23]–[25].

In contrast, standard PSI protocols based on Oblivious Pseudo-Random Functions (OPRF) [22] offer improved efficiency by reducing computational and communication overhead, making them particularly suitable for large datasets. These protocols allow the holder of the associated values to obtain the intersection and locally match the corresponding values, enabling further analysis.

However, the major limitation of OPRF-based protocols in the analysis is that the elements of $X \cap Y$ are disclosed to at least one party. Therefore, despite its efficiency, the OPRF-based PSI still requires privacy improvements in this case.

This work addresses the critical challenge of balancing efficiency and function adaptability while preserving individual

Corresponding author: Fan Zhang.

TABLE I
COMPARISON OF PSI VARIANTS AND PROTOCOLS

Protocol	Category	Functions Supported	Reveals Intersection	Computation / Communication Overhead
Circuit-PSI [4], [5], [9], [10]	General-purpose PSI	Arbitrary Functions	No	High / High
PSI-CA [6], [11]–[16]	PSI Variant	Cardinality	No	Medium / Low
PIS [17]–[19]	Task-specific	Sum	No	Medium / High
PSI-Stats [20]	Task-specific	Mean, Variance	No	Medium / High
PCSI [21]	Function-general	Cardinality, Sum, etc.	No	Medium / Medium
Approx-PSI (Ours)	Privacy-enhanced PSI Variant	Arbitrary Functions (Approximate)	Yes (but noised)	Low / Low

privacy in the analysis. We propose Approx-PSI, a novel two-party PSI variant designed for semi-honest adversaries. Built upon the highly efficient OPRF, the scheme enhances privacy protection by integrating the randomized response (RR) mechanism to achieve differential privacy (DP). It injects “noise” into the intersection, making it difficult for the receiver to definitively determine if an element truly belongs to the intersection or is merely a perturbation, thereby protecting individual privacy.

The contributions of this paper are listed as follows:

- *Differentially Private Set Intersection Protocol.* We developed a novel PSI in the semi-honest model, called Approx-PSI, using OPRF as a building block and innovatively applying a RR mechanism through two key strategies, *random obfuscation* and *membership conflict*. These strategies enable the receiver to learn a noised intersection and improve performance on large-scale datasets.
- *Approximate Intersection-based Analysis Framework.* We provide a theoretical framework to measure the similarity between the noised and the real intersection under different privacy parameters, ensuring that the receiver can perform any approximate analyses without compromising the individual privacy.
- *Experimental evaluation.* Our protocol achieves significant improvements in efficiency: the computation is at least $1.5 - 3\times$ (up to $77 - 81\times$) faster while reducing communication overhead by at least 90% compared to existing schemes. Moreover, we evaluate its adaptability using real-world datasets, and the results confirm that our scheme is highly practical for applications.

The remainder of this paper is organized as follows. Section II reviews related works, and Section III introduces preliminary concepts. Our proposed protocol is presented in Section IV, with detailed parameter configurations and strategies discussed in Section V. Section VI shows that the noised intersection remains useful for analysis while preserving privacy. The implementation details and evaluation results are provided in Section VII, followed by a comprehensive discussion of the findings in Section VIII. Finally, Section IX concludes the study.

II. RELATED WORK

While standard PSI protocols focus on securely learning the intersection between two parties, many applications require analyzing the intersection and its associated values. We refer to this emerging need as *intersection-based analysis*, which

generalizes PSI by supporting functions over the intersection. We summarize key representative schemes in Table I and review their methodologies, capabilities, and limitations below.

A. Standard PSI Protocols

Public-Key Encryption-based PSI [26] uses Diffie-Hellman key exchange or commutative encryption. This approach is conceptually simple and has low communication overhead but requires heavy computation. Homomorphic Encryption-based PSI [3], [12] enables one party to send an encrypted representation of its set and let the other perform intersection tests securely. This is especially attractive in unbalanced settings with very low communication, but is less practical for two datasets that are both similar and large.

C-PSI [4], [5] represents the intersection computation as a secure Boolean circuit evaluated through garbled-circuit protocols. It can easily support functionality beyond just revealing the intersection, but typically incurs higher computation and communication overhead.

OPRF-based PSI [1], [2], [22] provides the best balance of efficiency and security for large-scale datasets. Liang et al. [27] have provided a comprehensive performance benchmark for various PSI schemes, confirming that OPRF-based methods are often the most lightweight and scalable. In particular, fixed-key OPRF [7], [28], [29] allow for an efficient, unbalanced setting with lightweight clients.

Those standard PSI protocols disclose the intersection elements to at least one party and cannot preserve privacy when further analysis is conducted. Our work builds upon advances of OPRF-based PSI by introducing DP to mitigate privacy leakage in the intersection.

B. Intersection-based Analysis

C-PSI provides flexibility for securely evaluating complex functions over the intersection. Though the recently optimized [5], [10] have improved performance, they remain computationally expensive and are more applicable to small-scale datasets or scenarios requiring intricate logic evaluation.

PSI-CA schemes [15], [16] primarily output cardinality without revealing any elements of parties. Only cardinality is needed, so it is more efficient than standard PSI due to avoid expensive steps.

In addition, the PIS schemes [17]–[19] and PIS-Stats [20], a series of task-specific protocols, have been proposed to perform specific analyses, including the sum of associated values and the mean and variance. All these protocols rely heavily on homomorphic encryption to guarantee privacy,

which leads to expensive computation and communication costs.

The PCSI scheme [21], also a task-specific protocol, proposed a protocol to securely compute $g(X \cap Y)$ for a (mostly) generic choice of function g (like cardinality and sum in different settings), even used to privately compute the union. The PEPSI [30] also supports such functions, but it is optimized for highly unbalanced set sizes.

Nevertheless, revealing the intersection itself, its cardinality, or aggregates over associated values can still compromise privacy [23], [25]. Recent studies have combined DP with PSI (DP-PSI) to either protect aggregate outputs [13], [24], [31] or the intersection itself [14], [32], [33]. The latter is closest to ours but relying on heavy primitives.

Existing protocols' high computation and communication overheads, as well as limited support for diverse functions and potential privacy issues, limited their applicability in large-scale and privacy-sensitive settings. To overcome these challenges, we propose a lightweight DP-PSI framework that supports approximate intersection-based analysis to achieve a practical balance between performance, scalability, and privacy.

III. PRELIMINARIES

This section introduces some notations used when we describe our proposal, as well as the necessary background knowledge.

A. Notions and Security Model

We let κ, λ denote the computational and statistical security parameters, respectively. We write $[m]$ to denote a set $\{1, \dots, m\}$, and use $|A|$ to denote the size of sets.

Our proposed protocol implementation is based on the semi-honest adversary model. This is a standard assumption in security analysis, which describes an adversary who strictly follows the rules of the protocol but attempts to infer as much secret information as possible from the data they receive. Namely, such adversaries do not tamper with messages or deviate from the protocol, they fully exploit the legally exchanged information during protocol execution to disclose additional information.

Conversely, the malicious adversary model considers an opponent who not only tries to gather information from protocol interactions but also actively deviates from the protocol, such as sending fake messages, launching denial-of-service attacks, or exploiting vulnerabilities to directly assault the system for malicious purposes.

B. PSI and OPRF

The PSI is a specialized Secure Multi-Party Computation (SMPC) protocol that enables two parties to determine the shared elements in their datasets without revealing any other elements. We consider that the sender and receiver, denoted $\mathcal{S}$ and $\mathcal{R}$, hold a set Y with size n_y and X with size n_x respectively.

Algorithm 1 and Fig. 1a jointly illustrate the ideal functionality of PSI, denoted as $\mathcal{F}_{\text{PSI}}$. The functionality allows only one

Algorithm 1 PSI Ideal Functionality $\mathcal{F}_{\text{PSI}}$

PARAMETER:

- A Receiver $\mathcal{R}$ with set size n_x .
- A Sender $\mathcal{S}$ with set size n_y .

FUNCTIONALITY:

- Wait for input set $X = \{x_1, \dots, x_{n_x}\} \subseteq \{0, 1\}^*$ from $\mathcal{R}$.
- Wait for input set $Y = \{y_1, \dots, y_{n_y}\} \subseteq \{0, 1\}^*$ from $\mathcal{S}$.
- Compute the intersection $Z = X \cap Y$.
- Give output Z to $\mathcal{R}$.

participant, the receiver $\mathcal{R}$, to obtain the intersection. But there is still a problem that the $\mathcal{R}$ can learn from the intersection that an individual exists in the $\mathcal{S}$'s database. Therefore, we must redefine the functionality of PSI, incorporating DP and its application in our scenarios.

OPRF is also an SMPC protocol and an important building block. It enables one party to compute the output of a secret random function with respect to a given input value, while the other party, providing the function, remains oblivious to the specific input. OPRF-based PSI enables both parties to evaluate functions in a highly parallel manner, leading to typically low computational complexity and communication overhead.

C. Intersection-based Analysis

In intersection-based analysis, two parties, $\mathcal{S}$ and $\mathcal{R}$, collaborate to compute insights over the intersection of their datasets while preserving privacy. The $\mathcal{S}$ holds a set $Y = \{y_i\}_{i=1}^{n_y}$, where each y_i represents an element in the sender's dataset. The $\mathcal{R}$ holds a set $X = \{(x_i, t_i)\}_{i=1}^{n_x}$, where each x_i is an element, and t_i denotes the associated value linked to x_i .

The intersection is defined as $X \cap Y = \{x_i | x_i \in X, x_i \in Y\}$, with the associated values denoted as $\{t_i | x_i \in X \cap Y\}$. The objective is to perform a function, $F(X \cap Y)$, on the intersection and its associated values, where F can involve operations on the intersection, the values, or both. The result is returned to the receiver while ensuring that neither party learns any information about the elements of the intersection nor the elements outside the intersection..

D. Differential Privacy and Randomized Response

Differential privacy [34] is a rigorous mathematical framework designed to ensure privacy when sharing or publishing data. The key idea is that the output of any the query remains nearly the same, regardless of whether a specific individual's data is included, quantified by the privacy parameter ϵ . A smaller ϵ indicates stronger privacy protection, although it may compromise the data utility.

Formally, a randomized mechanism M is said to satisfy ϵ -DP if for any two neighboring datasets x and x' :

$$\Pr[M(x) \in S] \leq e^\epsilon \cdot \Pr[M(x') \in S]. \quad (1)$$

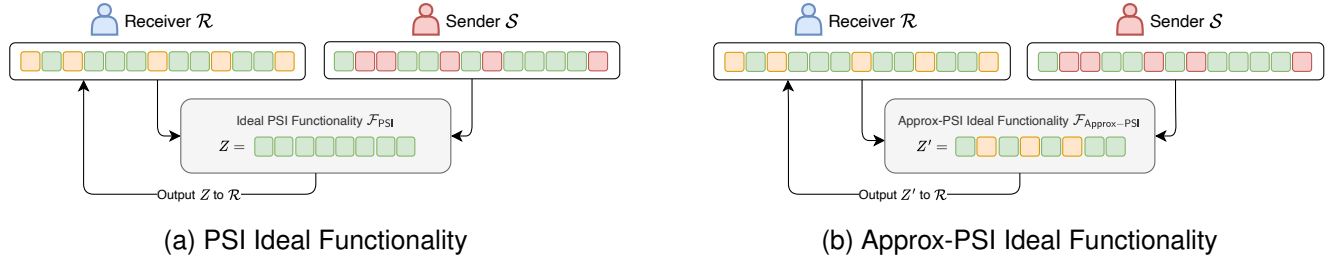


Fig. 1. Comparison of ideal functionalities in PSI and Approx-PSI, where green blocks represent elements of the real intersection, while yellow and red blocks indicate non-intersecting elements. (a) The standard PSI ideal functionality $\mathcal{F}_{\text{PSI}}$ returns the real intersection Z to the receiver $\mathcal{R}$. (b) The Approx-PSI ideal functionality $\mathcal{F}_{\text{Approx-PSI}}$ returns a differentially private noised intersection Z' .

The *post-processing* property of DP [35] ensures that any subsequent functions applied to the differentially private output maintain the privacy guarantees. Thus, if a mechanism M is ϵ -differentially private, then any function f applied to its output, $f(M)$, will also satisfy ϵ -DP.

Randomized response, introduced by Warner [36], enhances privacy in data collection by introducing randomness into the response process.

In the *generalized randomized response* mechanism, we denote the probability of responses as follows:

$$\Pr(X_i = k | x_i = j) = P_{jk}. \quad (2)$$

The RR mechanism is characterized by the following design matrix:

$$P = \begin{pmatrix} P_{00} & P_{01} \\ P_{10} & P_{11} \end{pmatrix}. \quad (3)$$

To ensure that the mechanism satisfies ϵ -DP, the following constraints must be met:

$$\begin{aligned} P_{11} &\leq e^\epsilon (1 - P_{00}) \\ P_{00} &\leq e^\epsilon (1 - P_{11}) \\ 1 - P_{00} &\leq e^\epsilon P_{11} \\ 1 - P_{11} &\leq e^\epsilon P_{00}. \end{aligned} \quad (4)$$

IV. PSI PROTOCOL BASED ON RANDOMIZED RESPONSE

In this section, we detail our proposed protocol, which integrates the RR mechanism to enhance privacy. We also provide a formal proof that our protocol is secure against a semi-honest adversary.

A. Definition

As described in Algorithm 2 and Fig. 1b, we propose a variant of the ideal functionality of the PSI protocol. At last $\mathcal{S}$ obtains nothing, while $\mathcal{R}$ obtains a *noised intersection* securely. The term “noised” indicates that $\mathcal{R}$ not only remains oblivious to elements outside the intersection but is also uncertain about the inclusion of certain elements from the intersection due to the inherent randomness of the protocol.

To clarify the distinction, we compare the results obtained by the functionalities $\mathcal{F}_{\text{PSI}}$ and $\mathcal{F}_{\text{Approx-PSI}}$. Let $Z = X \cap Y$ represents the *real intersection* in Algorithm 1, based on

Algorithm 2 Approx-PSI Ideal Functionality $\mathcal{F}_{\text{Approx-PSI}}$

INPUT:

- Receiver $\mathcal{R}$ holds a set $X = \{x_1, \dots, x_{n_x}\}$, where $x_i \subseteq \{0, 1\}^*$.
- Sender $\mathcal{S}$ holds a set $Y = \{y_1, \dots, y_{n_y}\}$, where $y_i \subseteq \{0, 1\}^*$.
- DP parameter $\epsilon > 0$.

FUNCTIONALITY:

- Wait for input set X from $\mathcal{R}$.
- Wait for input set Y from $\mathcal{S}$.
- Given output Z' to $\mathcal{R}$, where for each $x_i \in X, i \in [n_x]$:
 - If $x_i \in X \cap Y$, then the probability that x_i is included in Z' is $\Pr[y_i \in Z' | x_i \in X \cap Y]$.
 - If $x_i \in X \setminus Y$, then the probability that x_i is included in Z' is $\Pr[x_i \in Z' | x_i \in Y \setminus X]$.

traditional PSI functionality. In contrast, Z' in Algorithm 2 denotes the *noised intersection* produced by our protocol under DP.

Specifically, our mechanism ensures that for any subset $S \subset Y$, the probabilities $\Pr[Z' = S | x]$ and $\Pr[Z' = S | x']$ for all neighboring sets x and x' satisfy the DP constraint $\Pr[Z' = S | x] \leq e^\epsilon \cdot \Pr[Z' = S | x']$. This ensures that individual elements from X (or Y) cannot be identified with high confidence from the output Z' .

B. Protocol

The Approx-PSI protocol securely computes a differentially private noised intersection through four main phases, as illustrated in Fig. 2 and formalized in Algorithm 3. It involves two parties: a sender $\mathcal{S}$ and a receiver $\mathcal{R}$, with only $\mathcal{R}$ learning the output.

In the first stage, each party locally hashes their dataset using H_0 to uniformize the element representations. The $\mathcal{R}$ hashes its input set X to obtain $\{H_0(x_i)\}_{i \in [n_x]}$, and the $\mathcal{S}$ processes its set Y similarly.

In the second stage, the $\mathcal{S}$ perturbs its set using randomized obfuscation: each hashed element in $\{H_0(y_j)\}$ is independently removed or replaced with probability p_r , thereby making the inclusion of real intersection elements probabilistic.

In the third stage, both parties engage in a fixed-key OPRF protocol. The $\mathcal{R}$ obtains $T_i = F(k, H_0(x_i))$ for each $x_i \in X$,

Algorithm 3 The Approx-PSI Protocol

INPUT:

- Receiver $\mathcal{R}$ holds set $X = \{x_1, \dots, x_{n_x}\}$, where $x_i \in \{0, 1\}^*$ and $|X| = n_x$.
- Sender $\mathcal{S}$ possesses set $Y = \{y_1, \dots, y_{n_y}\}$, where $y_i \in \{0, 1\}^*$ and $|Y| = n_y$.

PARAMETERS:

- Random substitution probability p_r .
- Element conflict probability p_c .
- Hash function $H_0 : \{0, 1\}^* \rightarrow \{0, 1\}^\kappa$.
- Oblivious Pseudo-random Function (OPRF) $F : \{0, 1\}^* \times \{0, 1\}^* \rightarrow \{0, 1\}^{\lambda + \log(n_x n_y)}$.

FUNCTIONALITY:

- 1) Element Hashing:
 - Both $\mathcal{R}$ and $\mathcal{S}$ hash each element of their respective sets using H_0 : $\{H_0(x_i)\}_{i=1}^{n_x}$ and $\{H_0(y_i)\}_{i=1}^{n_y}$.
- 2) Random Obfuscation:
 - $\mathcal{S}$ randomly replaces or removes each element in $\{H_0(y_i)\}_{i=1}^{n_y}$ with a probability p_r , resulting in a modified set $\{H_0(y_i)'\}_{i \in [n'_y]}$.
- 3) Fixed-key OPRF Evaluation:
 - $\mathcal{R}$ and $\mathcal{S}$ engage in an Fixed-key OPRF instance F . $\mathcal{S}$ acts as the sender and receives a key k , while $\mathcal{R}$ acts as the receiver with input $\{H_0(x_i)\}$. For each element x_i , $\mathcal{R}$ receives $T_i = F(k, H_0(x_i))$. The Fixed-key OPRF ensures that $\mathcal{S}$ cannot learn any information about x_i beyond what is already included in T_i .
- 4) Membership Conflict:
 - $\mathcal{S}$ locally computes $\{Q_i\}_{i \in [n'_y]}$ where $Q_i = F(k, H_0(y_i)')$.
 - $\mathcal{S}$ initializes an approximate membership query data structure D , with a false positive probability p_c .
 - $\mathcal{S}$ sends $\mathcal{O} = \{D(Q_i)\}_{i \in [n'_y]}$ to $\mathcal{R}$.
 - $\mathcal{R}$ checks for each T_i whether $D(T_i)$ is present in $\mathcal{O}$. If $D(T_i) \in \mathcal{O}$, then corresponding x_i is included in the noised intersection $Z' = \{x_i \in X \mid D(T_i) \in \mathcal{O}\}_{i \in [n_x]}$.

while the $\mathcal{S}$ locally evaluates $Q_j = F(k, H_0(y'_j))$ over its obfuscated set using the same key k .

In the final stage, the $\mathcal{S}$ encodes $\{Q_j\}$ into an approximate membership query (AMQ) data structure D (Hash Table or a Bloom filter)—parameterized to achieve a target false positive rate p_c . The $\mathcal{S}$ then sends the representation $\mathcal{O}$ of D to the $\mathcal{R}$. The $\mathcal{R}$ performs approximate membership queries: for each T_i , it checks whether $D(T_i) \in \mathcal{O}$, forming the output noised intersection Z' .

To realize DP in this process, we design and introduce two key strategies: *random obfuscation* and *membership conflict*.

Random obfuscation prevents the $\mathcal{R}$ from reliably recovering real intersection elements by introducing randomness into the $\mathcal{S}$'s dataset. Specifically, elements in Y are randomly replaced or removed with probability p_r , so some true matches may be omitted. Since $\mathcal{S}$ does not know the real intersection, it cannot deliberately bias the result.

Membership conflict aims to obfuscate the uncertainty on the $\mathcal{R}$'s side by allowing approximate membership querying with controlled false positives. This is realized by encoding the perturbed elements into a data structure with probability p_c , allowing non-intersecting elements to be misidentified as matches, thereby further obscuring the output.

From $\mathcal{R}$'s perspective, the protocol can also be viewed as a randomized sampling process that partitions X into real intersection elements ($X \cap Y$) and non-intersection elements ($X \setminus Y$). These elements are sampled into 4 subsets according to specific probabilities:

- *True Positives (TP)* has real intersection elements which are included in the Z' with probability P_{TP} .
- *False Negatives (FN)* has real intersection elements which are excluded from the Z' with probability P_{FN} .
- *False Positives (FP)* has non-intersecting elements which are included in the Z' with probability P_{FP} .
- *True Negatives (TN)* has non-intersecting elements which are excluded from the Z' with probability P_{TN} .

The Z' is thus composed of elements from the *TP* and *FP* subsets. Furthermore, our protocol implements the RR mechanism as shown in (3), and the probabilities P_{TP} and P_{FP} correspond to the likelihoods $Pr[y_i \in Z' \mid x_i \in X \cap Y]$ and $Pr[x_i \in Z' \mid x_i \in Y \setminus X]$ in Algorithm 2, respectively. More importantly, neither party can learn the real intersection nor the elements in each subset. The probabilities are defined as follows:

$$\begin{aligned} P_{TP} &= (1 - p_r) + p_r p_c \\ P_{FN} &= p_r (1 - p_c) \\ P_{FP} &= p_c \\ P_{TN} &= 1 - p_c. \end{aligned} \tag{5}$$

C. Security

Let $\mathcal{S}$ (sender) hold set Y and $\mathcal{R}$ (receiver) hold set X . All “negligible” bounds are in the security parameter λ .

Theorem 1. *The protocol $\Pi_{\text{Approx-PSI}}$ in Algorithm 3 securely realizes the ideal functionality $\mathcal{F}_{\text{Approx-PSI}}$ depicted in Algorithm 2 in the semi-honest setting, assuming that the*

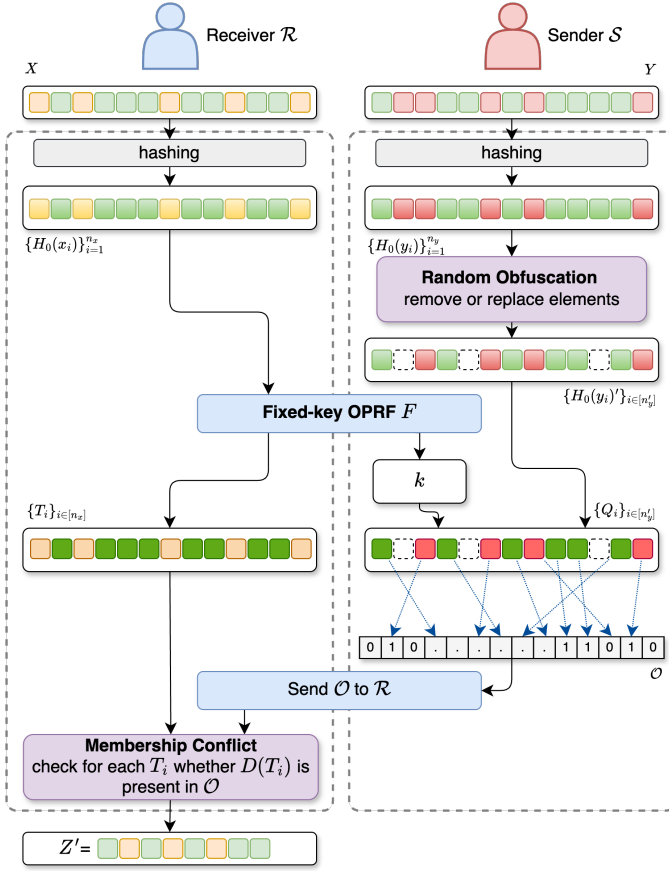


Fig. 2. An overview of Approx-PSI is presented, where green blocks denote elements of the real intersection, and the color changes represent different processes at different stages. In this protocol, the fixed-key OPRF is a building block, and the RR mechanism is primarily implemented via two strategies: random obfuscation and membership conflict. Finally, the S obtains nothing, and the R obtains a noised intersection.

oblivious pseudo-random function (OPRF) F is semi-honest and let H_0 is a collision-resistant hash function.

Proof: Let $\text{View}_{\mathcal{A}}^{\Pi}(X, Y)$ be the adversary's view in the real protocol execution with inputs (X, Y) , and let $\text{Ideal}_{\mathcal{F}, \text{Sim}_{\mathcal{A}}}(X, Y)$ denote the simulated view under the ideal functionality. We prove that for any probabilistic polynomial-time (PPT) semi-honest adversary $\mathcal{A}$, it holds that:

$$\text{View}_{\mathcal{A}}^{\Pi}(X, Y) \approx_c \text{Ideal}_{\mathcal{F}, \text{Sim}_{\mathcal{A}}}(X, Y), \quad (6)$$

where $\approx_c$ denotes computational indistinguishability. We denote $\text{Ideal}_{\mathcal{F}, \text{Sim}_{\mathcal{A}}}(X, Y)$ the simulated view produced by a simulator $\text{Sim}_{\mathcal{A}}$ interacting with the ideal functionality $\mathcal{F}$, given inputs (X, Y) . This models the ideal-world execution where the adversary learns only what is allowed by $\mathcal{F}_{\text{Approx-PSI}}$. Then we construct two simulators, depending on whether S or R is corrupted.

Case 1: Simulator Sim_S for corrupted sender.

The simulator Sim_S is given only the sender's input set Y . It simulates the view of S in the protocol as follows:

- 1) Select a random key $k \leftarrow \{0, 1\}^{\lambda}$ and whenever the adversary expects an OPRF response, return $F(k, \cdot)$. Because k is unknown to S and F is pseudorandom, these values are indistinguishable.

- 2) Generate a random approximate membership query data structure (e.g. Bloom filter) with p_c , independent of the R 's input.

Indistinguishability: As S does not receive any protocol output and only sees OPRF messages, which are simulatable, its view is indistinguishable from the real execution.

Case 2: Simulator Sim_R for corrupted receiver.

The simulator Sim_R is given only R 's input X and the output Z' from $\mathcal{F}_{\text{Approx-PSI}}$. It simulates the view of R in the protocol as follows:

- 1) Sample a fresh key $k \leftarrow \{0, 1\}^{\lambda}$ and answer every future query on $H_0(x)$ with $\tilde{T}_x := F(k, H_0(x))$.
- 2) For each $x \in Z'$ compute $\tilde{T}_x$ and add it to Q_1 .
- 3) For every $x \in X \setminus Z'$ flip an independent coin $b_x \sim \text{Bernoulli}(p_c)$. If $b_x = 1$ add $\tilde{T}_x$ to Q_0 .
- 4) Build the AMQ structure $\mathcal{D}$ over $Q = Q_1 \cup Q_0$ using the same public parameters as the protocol, such as the bit array size and the number of hash functions in the Bloom filter. Compute the message $\mathcal{O} := \{\mathcal{D}(T) \mid T \in Q\}$ and send $\mathcal{O}$ to the adversary $\mathcal{A}$.
- 5) Respond to any new queries from $\mathcal{A}$ with the oracle defined in Step 1.

Indistinguishability. (i) The multiset $\{\tilde{T}_x\}_{x \in X}$ in both worlds are generated under a random hidden key, thus identical. (ii) For every $x \in X \setminus Z'$ each membership bit $\mathcal{D}(\tilde{T}_x)$ equals 1 with the same independent probability p_c in both worlds; hence the multiset $\mathcal{O}$ has an identical distribution. (iii) Any distinguishing event requires a hash collision in H_0 , which occurs with negligible probability.

Therefore, for every PPT distinguisher $\mathcal{D}$,

$$|\Pr[\mathcal{D}(\text{View}^{\Pi}) = 1] - \Pr[\mathcal{D}(\text{Ideal}) = 1]| \leq \text{negl}(\lambda). \quad (7)$$

Hence $\Pi_{\text{Approx-PSI}}$ securely realizes $\mathcal{F}_{\text{Approx-PSI}}$ in the semi-honest setting under the stated assumptions.

V. SETTINGS AND STRATEGIES

In this section, we detail the parameter settings and strategies of our protocol, focusing on balancing accuracy and privacy by carefully selecting the privacy parameter ϵ .

A. Privacy Parameters and Settings

As our proposal is expected to balance accuracy and privacy, we only aim to minimize P_{FP} and P_{FN} probabilities while satisfying the RR constraints shown in (4). These constraints are defined as:

$$\begin{aligned} P_{TP} &\leq e^{\epsilon} P_{FP} \\ P_{TN} &\leq e^{\epsilon} P_{FN}. \end{aligned} \quad (8)$$

From these constraints and the relationships between the probabilities in (8) and (5), we derive the following conditions:

$$\frac{P_{TN}}{P_{FN}} = \frac{1 - p_c}{p_r(1 - p_c)} = \frac{1}{p_r} \leq e^{\epsilon} \Rightarrow p_r \geq e^{-\epsilon}, \quad (9)$$

$$\frac{P_{TP}}{P_{FP}} = \frac{(1 - p_r) + p_r p_c}{p_c} = p_r + \frac{(1 - p_r)}{p_c} \leq e^{\epsilon}. \quad (10)$$

To maximize accuracy, we set $p_r = e^{-\epsilon}$. Substituting this into the second equation yields $p_c \geq (1 + e^\epsilon)^{-1}$, so we set $p_c = (1 + e^\epsilon)^{-1}$ for optimal accuracy. Thus, we define:

$$\begin{aligned} p_r &= e^{-\epsilon}, \\ p_c &= \frac{1}{1 + e^\epsilon}. \end{aligned} \quad (11)$$

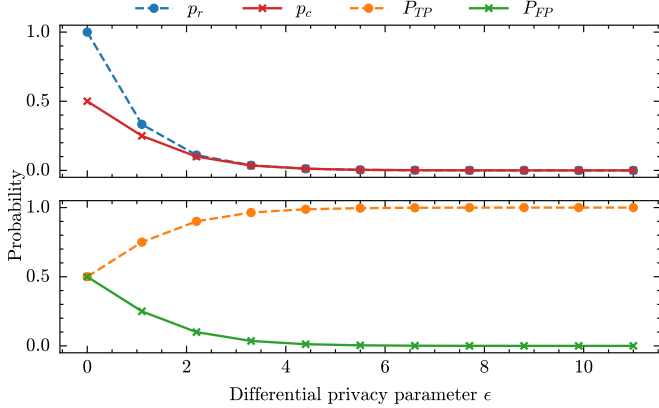


Fig. 3. Probabilities of random obfuscation (p_r), membership conflict (p_c), true positive (P_{TP}), and false positive (P_{FP}) vary with the differential privacy parameter (ϵ). As ϵ increases, p_r and p_c decrease, reducing the noise in the intersection. This weakens privacy protection, as less noise increases the risk of inferring individuals.

Fig. 3 illustrates how p_r and p_c depend solely on the DP parameter ϵ , with no relation to the size of the sets X and Y or the real intersection.

As ϵ increases, the probability of random obfuscation (p_r) decreases, leading to fewer elements being randomly substituted or removed. For large ϵ , p_r approaches zero, indicating that random obfuscation is no longer applied. Similarly, the probability of membership conflict (p_c) decreases with ϵ .

B. Random Obfuscation

Random obfuscation is utilized to modify the real intersection by Random Replace or Random Remove elements within the set Y probabilistically. Specifically, the obfuscation can either:

- Random Replace: Replace elements in Y with random strings $r \in \{0, 1\}^{\kappa}$, which keeps the size of Y unchanged.
- Random Remove: Remove elements from Y , which reduces the size of Y to n'_y .

Obviously, Random Replace maintains the size of Y , while Random Remove might reduce computational and communication costs by decreasing the size of Y depending on p_r . When $p_r = 0$, $n'_y = n_y$ (no obfuscation), and when $p_r = 1$, n'_y can be closely reduced to 0 (complete obfuscation). This makes Random Remove preferable in performance-sensitive scenarios theoretically. When ϵ is small and p_r is large, a significant portion of Y is removed, resulting in a more optimized performance.

C. Membership Conflict

Membership conflict addresses the issue of falsely matched non-intersection elements between sets X and Y . This step uses *approximate membership queries*, with two optional implementations: Hash Table and Bloom Filter.

For Hash Table, this probability is closely related to its length, denoted as N . A suitable length N will lead to a certain probability of hash collision between two different elements, which can meet the needs of the strategy.

A hash collision occurs when different items are hashed into the same bin. The probability of two items colliding equals $1/N$ if the hash is uniform. Overall, if the size of elements expected to be stored is n (equivalent to the size of the sender's datasets n_y), the probability of a membership conflict of the query is:

$$p_c = 1 - \left(1 - \frac{1}{N}\right)^n. \quad (12)$$

Thus, we can control the p_c by adjusting N , when n is fixed. We substitute (12) into (11) and rewrite it as:

$$\begin{aligned} p_c &= 1 - \left(1 - \frac{1}{N}\right)^n \geq \frac{1}{1 + e^\epsilon}, \\ &\Rightarrow \frac{1}{N} \geq 1 - \left(1 - \frac{1}{1 + e^\epsilon}\right)^{\frac{1}{n}}, \\ &\Rightarrow \frac{1}{N} \geq 1 - \left(\frac{e^\epsilon}{1 + e^\epsilon}\right)^{\frac{1}{n}}, \\ &\Rightarrow \frac{n}{N} \geq n \left[1 - \left(\frac{e^\epsilon}{1 + e^\epsilon}\right)^{\frac{1}{n}}\right]. \end{aligned} \quad (13)$$

The right part in (13) tends to be a fixed value as n grows when ϵ is fixed. Then we can get:

$$\begin{aligned} \lim_{n \rightarrow \infty} \left\{ n \left[1 - \left(\frac{e^\epsilon}{1 + e^\epsilon} \right)^{\frac{1}{n}} \right] \right\} &= -\ln \left(\frac{e^\epsilon}{1 + e^\epsilon} \right) \\ &= \ln(1 + e^{-\epsilon}). \end{aligned} \quad (14)$$

Thus, we slightly strengthen the constraint from (13) to (15) as follows:

$$\frac{n}{N} \geq \ln(1 + e^{-\epsilon}). \quad (15)$$

Then we can get $N \leq n / \ln(1 + e^{-\epsilon})$. When n is fixed, by specifying the DP parameters, the bit array length required to implement the strategy through the Hash Table can be determined.

Similarly, Bloom Filter can meet the requirement, which is a highly space-efficient probabilistic data structure. It is primarily designed to check whether an element might be a member of a set with a non-zero false positive probability, and may mistakenly report an element as present in the set.

The probability of false positive of Bloom Filter can be controlled by adjusting several parameters, including the length of the bit array, the number of hash functions used, and the expected size of elements to be inserted. Here, n represents the size of elements inserted into the Bloom Filter, and M

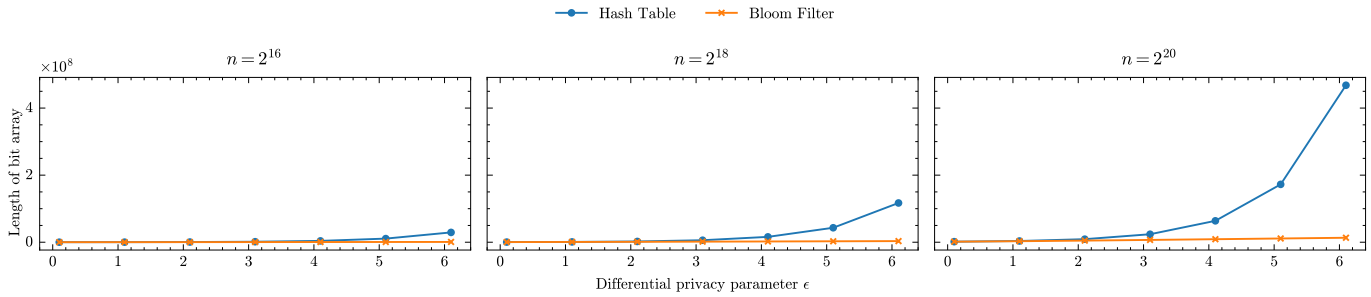


Fig. 4. Bit array length of Hash Table and Bloom Filter under various scale settings with DP parameter (ϵ). Compared to the Bloom Filter, the Hash Table requires significantly more bits as the dataset size or ϵ increases.

denotes the length of the bit array used in the filter. The k refers to the size of hash functions each element is mapped to in the bit array.

The probability of membership conflict of Bloom Filters is given by (16) as follows:

$$p_c = \left(1 - \left(1 - \frac{1}{M}\right)^{kn}\right)^k \leq \frac{1}{1 + e^\epsilon}. \quad (16)$$

Similarly, when M is extremely larger, we can approximate and simplify into:

$$\begin{aligned} \lim_{M \rightarrow \infty} \left(1 - \left(1 - \frac{1}{M}\right)^{kn}\right)^k &= (1 - e^{-\frac{kn}{M}})^k \\ &= \left(e^{-\frac{kn}{M}}\right)^k \\ &= e^{-\frac{k^2 n}{M}}. \end{aligned} \quad (17)$$

Thus, we can rewrite the (16) as follows:

$$\begin{aligned} e^{-\frac{k^2 n}{M}} &\leq \frac{1}{1 + e^\epsilon}, \\ \Rightarrow \frac{k^2 n}{M} &\leq \ln(1 + e^\epsilon), \\ \Rightarrow M &\leq \frac{k^2 n}{\ln(1 + e^\epsilon)}. \end{aligned} \quad (18)$$

In practical applications, in order to minimize the probability of false positive, p_c , the optimal size of hash functions k is usually taken as:

$$k = \frac{M}{n} \ln(2). \quad (19)$$

Based on (18) and (19), the relationship between the implementation parameters of the Bloom Filter and the DP parameters can be derived straightforwardly:

$$M \geq \frac{n \ln(1 + e^\epsilon)}{(\ln(2))^2}. \quad (20)$$

As shown in Fig. 4, Bloom Filter requires fewer bits than Hash Table for large datasets theoretically, making them more efficient in space. This efficiency grows as ϵ and the dataset size increase, demonstrating Bloom Filter's advantage in space utilization under DP constraints.

In summary, we have established clear relationships between the privacy parameters and the optimal settings for p_r and p_c , effectively managing the trade-offs between privacy and accuracy.

VI. APPROXIMATE ANALYSIS ON NOISED INTERSECTION

This section examines the similarity focusing on the influence of the parameter ϵ and other variables on the similarity between the noised and real intersections.

A. Similarity between Noised and Real Intersection

The sender uses random obfuscation to probabilistically exclude elements from the real intersection Z from appearing in the Z' . The exclusion probability p_r is directly influenced by ϵ , where a higher value of ϵ reduces the noise, allowing more real elements to be retained.

Similarly, membership conflict introduces false positives into the noised intersection. A higher membership conflict probability p_c , determined by ϵ , leads to more elements from outside Z entering Z' . Thus, the similarity between the real and noised intersection is jointly determined by p_r and p_c , both of which are affected by ϵ .

We denote π as the proportion of all elements X in $\mathcal{R}$ to the real intersection, where $\pi = |Z|/|X|$. It should be noted that π is merely a theoretical construction to our discussion and does not manifest within the actual protocol scheme.

For each element of the real intersection, it is probabilistically assigned to either the true positive (TP) or false negative (FN) subset, with respective probabilities P_{TP} and $P_{FN} = 1 - P_{TP}$. This categorization follows a Bernoulli distribution, denoted $Bern(P_{TP})$. Repeating this process $n_x \pi$ times, we model the size of true positives $|TP|$ as a random variable following a binomial distribution $B(n_x \pi, P_{TP})$. The distributions of $|FP|$, as well as its corresponding expected value and variance, can be derived similarly. Therefore, through the mechanism, the size of each subset, $|TP|$ and $|FP|$, follow binomial distributions with respective expected values and variances:

$$\begin{aligned} \mathbb{E}[|TP|] &= n_x \pi P_{TP}, \\ \mathbb{E}[|FP|] &= n_x (1 - \pi) P_{FP}, \end{aligned} \quad (21)$$

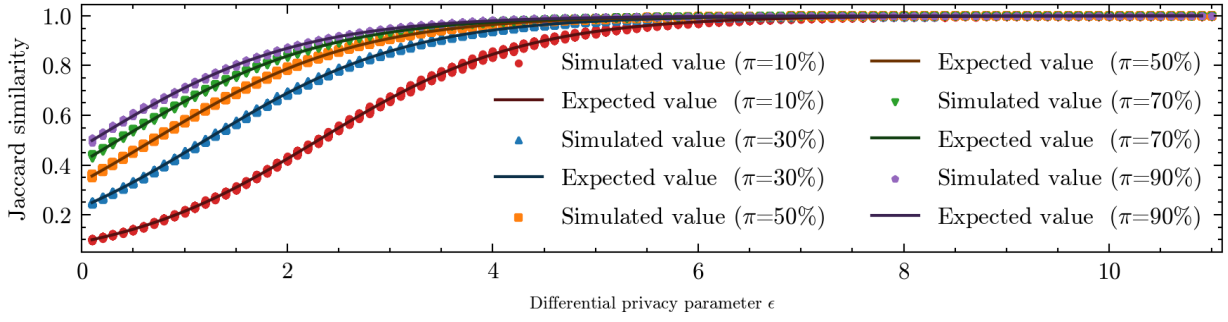


Fig. 5. The relationship between jaccard similarity and the differential privacy parameter ϵ is illustrated on varying proportions of the real intersection, where results are evaluated over 1000 times independent runs for each ϵ . The close alignment between the simulated and expected values confirms the correctness of (25). Additionally, this illustration highlights that there exists a practical range for ϵ wherein an acceptable balance between privacy protection and data utility can be achieved.

$$\begin{aligned} \text{Var}(|TP|) &= n_x \pi P_{TP} (1 - P_{TP}) \\ &= n_x \pi P_{TP} P_{FN}, \\ \text{Var}(|FP|) &= n_x (1 - \pi) P_{FP} (1 - P_{FP}) \\ &= n_x (1 - \pi) P_{FP} P_{TN}. \end{aligned} \quad (22)$$

We can observe that despite both P_{TP} and P_{FP} are affected by ϵ , these represent two distinct and independent random processes. These processes draw elements from non-overlapping subsets, specifically $X \cap Y$ and $X \setminus Y$, respectively. This independence implies that the random variables $|TP|$ and $|FP|$, representing the sizes of true positives and false positives, are also independent.

To evaluate the feasibility of the Z' as a substitute for the Z , we analyze the similarity between Z' and Z . The *jaccard similarity* is used to measure the similarity between the two results. Its value ranges from 0 to 1, if the two results are identical, the similarity is 1 due to they are equal. It can be expressed as:

$$J(Z, Z') = \frac{|Z \cap Z'|}{|Z \cup Z'|}. \quad (23)$$

From (23), we can simply observe that $|Z \cap Z'|$ is $|TP|$, and $|Z \cup Z'|$ is $|TP| + |FP| + |FN|$. Hence, we know that the expected value of the similarity between the Z' and the real intersection Z is:

$$\mathbb{E}[J(Z, Z')] \approx \frac{\mathbb{E}[|TP|]}{\mathbb{E}[|TP|] + \mathbb{E}[|FP|] + \mathbb{E}[|FN|]}. \quad (24)$$

From (21), it can be further simplified to:

$$\mathbb{E}[J(Z, Z')] \approx \frac{\pi P_{TP}}{\pi + P_{FP} - \pi P_{FP}}. \quad (25)$$

In our analysis, to simplify $\mathbb{E}[J(Z, Z')]$, we independently evaluated $\mathbb{E}[|TP|]$ and $\mathbb{E}[|FP|]$ before substituting them into the equation. This approach is justified due to limited interdependencies among TP and FP in our specific context, despite their shared stochastic parameters.

Combined with (5), we can know that $\mathbb{E}[J(Z, Z')]$ is directly related to ϵ . Therefore, when ϵ is large enough, the noise gradually decreases, and eventually the elements in Z

and Z' will be exactly the same. If the individual elements in the noised and the real gradually become consistent, their statistical characteristics (such as mean, variance, etc.) will also tend to be the same, as shown in the following:

$$\lim_{\epsilon \rightarrow \infty} \mathbb{E}[J(Z, Z')] = 1. \quad (26)$$

As shown in Fig. 5, we can observe that the similarity between the noised and the real is jointly affected by π and ϵ . The similarity is independent of the dataset size (for both $\mathcal{S}$ and $\mathcal{R}$), ensuring stability across different sizes when other parameters remain unchanged. Furthermore, the value of π significantly affects the similarity. Under the same DP level, a lower π results in lower similarity. However, different π will eventually cause the similarity to be close to 1 for a given ϵ . Since neither $\mathcal{S}$ nor $\mathcal{R}$ can know π in advance, we can only focus on how changes in ϵ affect similarity.

B. Approximate Intersection-based Analysis

Crucially, thanks to the post-processing property, this property ensures that any analysis or transformation applied to Z' does not further increase privacy risks.

Although Z' has been introduced with randomness noise, the data utility is preserved significantly. By carefully configuring ϵ , the Z' retains key characteristics of the real intersection. In addition, after obtaining Z' , the analysis functions over the associated values can be further generalized, independent of the value types or the number of associated values in the receiver's set.

In conclusion, Z' indeed reduces the accuracy of the analysis results. However, it can bring more advantages:

- *Privacy-preserving*: Only one party obtains Z' , and the other controlling ϵ learns nothing about Z .
- *Function Adaptability*: $\mathcal{R}$ can apply any analysis functions to Z' without additional privacy loss, enabling analyses beyond basic statistics.
- *Associated values Feasibility*: Z' supports multiple associated values in $\mathcal{R}$'s dataset without degrading performance or increasing privacy loss.

VII. EXPERIMENTAL EVALUATION

In this section, we rigorously evaluate the proposed protocol in terms of both functionality and performance.

A. Implementation Details

TABLE II
SUMMARY OF KEY EXPERIMENTAL PARAMETERS

Parameter	Setting
CPU	Intel i7-9700 @ 4.7GHz
Memory / OS	32 GB RAM, Arch Linux
Network Configuration	LAN (700 Mbps), WAN (50 Mbps); latency: 400ms
Generated Dataset Size ($ X = Y $)	$2^{14}, 2^{16}, 2^{18}, 2^{20}$
Real-world Datasets (Size) ($ X = Y $)	Adult(48,842), Bank(41,188), Sepsis(110,204)
Payload Value Type	Uniform integers (synthetic); real-world values
Threads	Parallel (offline); Single-threaded (online phase)

As described in Table II, we implement our protocol and all baseline methods in Java (based on `mpc4j`¹), ensuring consistency in the performance evaluation. For our protocol, we enable optimization in the offline phase via Java's built-in parallel stream mechanism. The online communication phase is executed in a single-threaded manner to ensure stable and deterministic behavior.

Both the sender and receiver sides are developed separately, and we leverage the fixed-key OPRF implementation from [29]. Moreover, we use computational and statistical security parameters $\kappa = 128$ and $\lambda = 40$.

All experiments are conducted on a machine running Arch Linux with an Intel i7-9700 CPU (4.7 GHz) and 32 GB of RAM. We simulate both LAN and WAN environments using `Traffic Control` to adjust the network conditions, where the LAN has a bandwidth of 700 Mbps, while the WAN has a bandwidth of 50 Mbps, both with a burst size of 32 kbit and a latency of 400 ms. We run experiments by simultaneously launching the receiver and sender on the same machine.

For the experiments, we generate dataset sizes of $2^{14}, 2^{16}, 2^{18}, 2^{20}$ for both the sender and the receiver, and the associated values in the receiver are generated uniformly distributed. Due to the constraints of some protocols, which only support positive integer payloads, we rounded the values to positive integers to ensure compatibility.

We also used datasets in the real-world for evaluation. In these experiments, $\mathcal{R}$ holds identifiers and values of a public dataset, and $\mathcal{S}$ holds part of identifiers of the public dataset and randomly generates a part of identifiers as a supplement. For the associated value, the values in the public dataset are used directly.

All communication and runtime metrics are reported in Megabytes (MB) and seconds (s), respectively. For the generated datasets, we compare the real intersection results with those obtained by the evaluated protocols. For the real-world datasets, we only report the results of our protocol.

B. Strategies

The experiment is conducted in a LAN environment, with a generated dataset size of 2^{20} both in $\mathcal{S}$ and $\mathcal{R}$. We evaluated

two randomized obfuscation strategies: Random Replace and Random Remove, alongside two membership conflict strategies: Hash Table and Bloom Filter. Each combination is tested under different privacy parameters ($\epsilon = \{1, 5, 10\}$).

Table III summarizes the performance comparison of different strategies for random obfuscation and membership conflict. Firstly, the results indicate that the communication for $\mathcal{R}$ remains constant across all strategies, primarily due to the fixed-key OPRF phase dominating its communication costs.

Random Replace vs. Random Remove: The primary difference lies in runtime performance. The remove strategy shows better performance, especially with lower ϵ . When ϵ is small, the probability p_r of removing elements is higher, leading to fewer elements in $\mathcal{S}$ and thus improved performance. However, as ϵ increases, the advantage of remove diminishes due to the decreasing probability of element removal.

Hash Table vs. Bloom Filter: The main distinction lies in the communication overhead. As ϵ increases, Hash Table exhibits higher communication costs, consistent with the expected bit array lengths shown in Fig. 4. In terms of runtime, the Bloom Filter outperforms the Hash Table slightly. These improvements are especially valuable in resource-constrained or latency-sensitive environments.

The combination of **Random Remove** for obfuscation and **Bloom Filter** for membership conflict consistently outperforms other configurations, which makes it the most efficient strategy in terms of both communication overhead and running time. Therefore, we adopt this combination for subsequent evaluations.

C. FPR and FNR

This evaluation presents an analysis of the effects of the privacy parameter and dataset size on key performance metrics, including true positives ($|TP|$), false positives ($|FP|$), false positive rate (FPR), and false negative rate (FNR).

The analysis covers multiple dataset sizes ($2^{14}, 2^{16}$, and 2^{18}) to analyze scalability. Each ϵ is evaluated over 1000 independent experimental runs, and the results are averaged and logarithmic transformation (base 10) to capture the overall

TABLE III
COMMUNICATION (MB) AND RUNNING TIME (S) IN LAN FOR DIFFERENT STRATEGIES

Strategies		ϵ	Communication		Running time	
			$\mathcal{S}$	$\mathcal{R}$	$\mathcal{S}$	$\mathcal{R}$
Replace	Hash Table	1	33.40	33.00	56.95	57.15
		5	51.61	33.00	57.08	57.51
		10	289.00	33.00	56.27	62.25
Remove	Hash Table	1	33.25	33.00	47.73	50.68
		5	51.49	33.00	56.06	56.53
		10	289.00	33.00	55.53	60.01
Replace	Bloom filter	1	33.34	33.00	50.25	57.84
		5	34.30	33.00	50.11	50.75
		10	35.60	33.00	50.76	51.19
Remove	Bloom filter	1	33.22	33.00	42.05	44.74
		5	34.29	33.00	49.41	49.70
		10	35.60	33.00	49.83	50.51

¹<https://github.com/alibaba-edu/mpc4j>

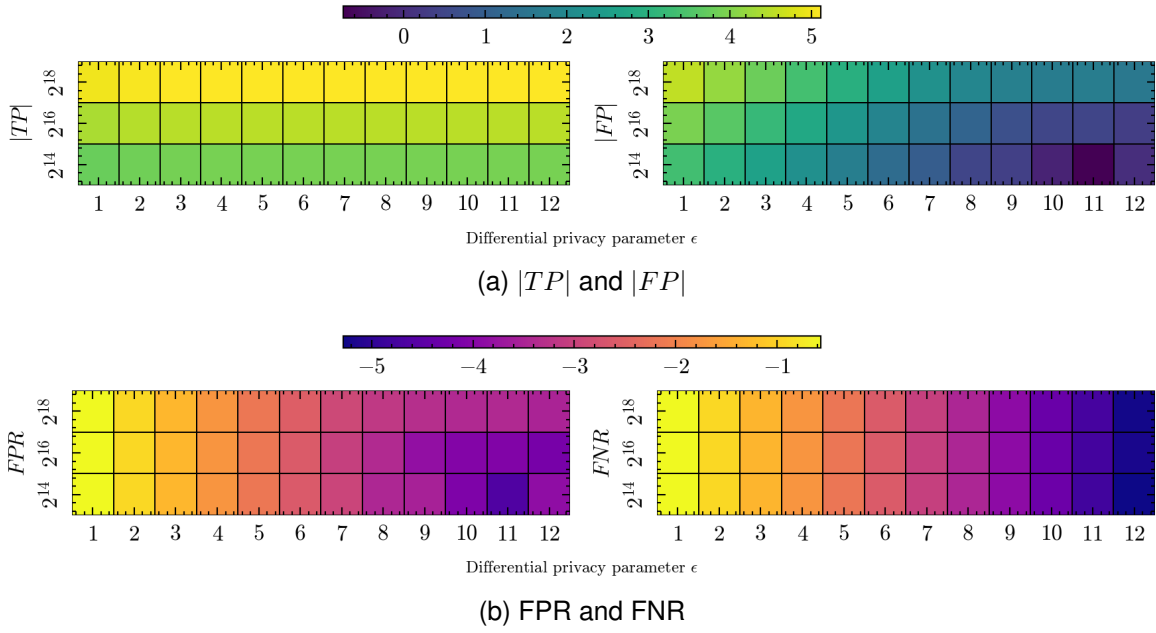


Fig. 6. Impact of dataset size and differential privacy parameter ϵ on (a) the sizes of true positives ($|TP|$) and false positives ($|FP|$), and (b) the rates of false positive (FPR) and false negative (FNR). Results are averaged over 1000 independent runs, with a logarithmic transformation (base 10).

trends and enhance the visibility of small but significant variations.

There are deterministic relationships between the sizes of the subsets, where $|TP| + |FN| = |X \cap Y|$, $|FP| + |TN| = |X \setminus Y|$. Given these relationships, we can focus on analyzing $|FP|$ and $|TP|$. Additionally, based on the concepts, we have:

$$\begin{aligned} FPR &= \frac{|FP|}{|FP| + |TN|}, \\ FNR &= \frac{|FN|}{|TP| + |FN|}. \end{aligned} \quad (27)$$

Privacy Parameter: As expected and shown in Fig. 6, the analysis reveals that smaller ϵ values, which indicate stronger privacy protection, lead to increased noise in the data, resulting in higher $|FP|$ and $|FN|$. Conversely, larger ϵ values produce a Z' that closely resembles the actual Z , reducing both FPR and FNR.

Dataset Size: As the dataset size increases, both $|TP|$ and $|FP|$ also grow. Furthermore, since FNR is a probabilistic metric, it remains unaffected by the size of the dataset, exhibiting stable behavior across datasets of varying scales. In contrast, $|FP|$ and FPR show greater variability with larger datasets, with their values increasing as the dataset size grows. This can be attributed to the increase in both the FPR and the $|FP|$ introduced by the Bloom Filter as the dataset size expands. Consequently, non-intersecting elements are more likely to be incorrectly matched as part of Z' , which amplifies the variability in the analytical results.

Overall, the effect of privacy parameter ϵ can be manageable and interpreted by changes of confusion in each subset.

D. Performance

The C-PSI [5] can perform arbitrary functions but requires specific settings, so we only evaluate it on cardinality. The

PCSI [21] is mainly based on the comprehensive application of multiple technologies, such as Oblivious Transfer and OPRF. It can perform the sum function but also requires a specific setting, so only evaluate it on size. In addition, it can be extended to use t_i^2 as the associated value to perform the mean or variance, but it requires multiple executions. The PIS [18] and PSI-Stats [20] are both mainly based on homomorphic encryption, whereas the PIS can perform size and sum. The PSI-Stats can perform size and mean, and it can be extended to t_i^2 as the associated value to perform the variance as well, but it requires more communication and computation.

For each scheme, we record the communication overhead and running time for both sender and receiver, and measured the running time in both network environments. We primarily present the results in LAN here for clarity, because the results of the WAN network are similar (The results of other schemes are deterministic, whereas the results of our protocol introduce slight variations due to noise). Additionally, the bold values in the Table IV are the actual values of the real intersection. Our scheme is evaluated using three privacy parameters ($\epsilon = \{1, 4, 6\}$), which are suitable for different levels of privacy protection in practical applications.

As presented in Table IV, our protocol demonstrates significant reductions in both communication and computational overhead when compared to existing schemes.

Communication Overhead: Our protocol achieves a reduction of at least 90% across all dataset sizes.

Running Time: Under LAN and WAN settings, our protocol achieves at least $1.5\times$ and up to $3\times$ faster performance than competing schemes, respectively. Notably, when compared to Google's scheme [18], we observe performance improvements of up to $81\times$ in LAN and $77\times$ in WAN, especially evident with larger datasets.

TABLE IV
RUNNING TIME (S), COMMUNICATION (MB) AND RESULTS FOR DIFFERENT SCHEMES

n	Protocol	ϵ	Communication			Running time (LAN)		Running time (WAN)		Result (LAN)			
			$\mathcal{S}$	$\mathcal{R}$	Total	$\mathcal{S}$	$\mathcal{R}$	$\mathcal{S}$	$\mathcal{R}$	Cardinality	Mean	Variance	Sum
2^{16}										32768	49.451	102.562	1620406
	C-PSI [5]	-	58.81	26.00	84.80	12.93	12.96	26.60	26.73	32768	-	-	-
	PCSI [21]	-	23.20	21.48	44.68	5.615	5.643	12.055	12.193	32768	-	-	-
	PIS [18]	-	2.06	53.12	55.19	305.21	305.22	312.46	312.47	32768	-	-	1620406
	PSI-Stats [20]	-	2.06	53.12	55.19	299.84	299.83	312.89	312.91	32768	49	-	-
	Ours	1	2.08	2.06	4.14	3.11	3.19	3.25	3.66	32793	49.516	103.077	1623762
		4	2.13	2.06	4.19	3.52	3.57	3.75	3.93	32845	49.453	102.593	1624273
		6	2.16	2.06	4.22	3.69	3.74	4.01	4.10	32764	49.454	102.553	1620317
2^{18}										131072	49.506	102.021	6488825
	C-PSI [5]	-	235.25	104.3	339.55	63.43	63.53	116.98	117.49	131072	-	-	-
	PCSI [21]	-	102.70	90.11	192.81	23.82	23.81	50.14	50.67	131072	-	-	-
	PIS [18]	-	8.25	212.50	220.75	1234.98	1234.99	1289.35	1289.36	131072	-	-	6488825
	PSI-Stats [20]	-	8.25	212.50	220.75	1204.94	1204.95	1252.68	1252.69	131072	49	-	-
	Ours	1	8.30	8.25	16.55	10.49	10.91	10.32	12.22	131077	49.525	101.965	6491560
		4	8.51	8.25	16.76	11.71	11.84	12.24	13.02	131154	49.514	102.055	6493918
		6	8.64	8.25	16.89	11.69	11.84	12.59	13.18	131136	49.506	102.037	6491960
2^{20}										524288	49.479	102.403	25940997
	C-PSI [5]	-	1002.33	440.37	1442.71	479.15	479.49	700.59	702.58	524288	-	-	-
	PCSI [21]	-	450.40	376.70	827.10	87.41	87.54	212.79	215.12	524288	-	-	-
	PIS [18]	-	33.00	850.00	883.00	4794.53	4794.54	4959.06	4959.07	524288	-	-	25940997
	PSI-Stats [20]	-	33.00	850.08	883.08	4813.99	4814.00	4931.85	4931.86	524288	49	-	-
	Ours	1	33.22	33.00	66.22	39.34	42.07	38.01	45.75	524465	49.492	102.467	25957016
		4	34.03	33.00	67.03	46.12	47.49	46.72	50.43	524922	49.481	102.396	25973521
		6	34.56	33.00	67.56	46.15	46.62	47.84	50.80	525219	49.480	102.399	25987642

E. Public Datasets

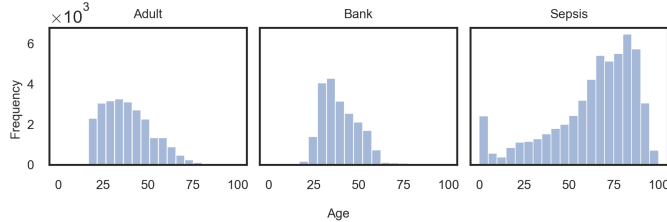


Fig. 7. Histogram of “age” distribution in the intersection of different public datasets (Adult, Bank, and Sepsis).

To evaluate the adaptability of our protocol in real-world scenarios, we applied it to three widely used public datasets from the UCI repository: Adult [37], Bank Marketing [38], and Sepsis [39].

We selected the three datasets with sizes of 48,842 (Adult), 41,188 (Bank Marketing), and 110,204 (Sepsis) records, focusing on the ‘age’ variable as the associated value due to its varied distribution characteristics, as shown in Fig. 7. The statistical results for intersection size, mean, variance, and sum were computed under different ϵ values.

As expected and the results presented in Fig. 8, lower ϵ values introduce more noise, which adversely affects the accuracy of the statistical measures. Conversely, as ϵ increases, the impact of noise diminishes, leading to results that align closely with the actual dataset values.

Notably, the findings affirm that our scheme is adaptable to a wide range of datasets, effectively managing privacy concerns without significantly compromising the accuracy of statistical results.

VIII. DISCUSSION

In this section, we outline the key aspects of the proposed scheme and discuss its implications.

A. Balancing Privacy and Utility

Our experimental results show that the impact of noise varies based on the different analysis functions. Some metrics such as the mean retain high utility even with moderate noise. However, size and sum exhibit greater sensitivity to noise, necessitating a careful balance between privacy and analytical accuracy.

Notably, as dataset size increases, the relative impact of noise diminishes, not merely because larger datasets dilute probabilistic noise, but also due to the natural obfuscation effect created by the complexity and overlap of data. As the size of intersecting and non-intersecting elements increases, distinguishing individual elements becomes inherently more challenging, enhancing privacy protection even with reduced noise. This implies that larger datasets inherently enhance privacy, allowing for a more flexible privacy budget design without compromising data security. Therefore, dataset size emerges as not only a technical parameter but also a strategic factor in balancing privacy and utility, driving both higher the analytical accuracy and improved scalability.

Additionally, our use of optimized data structures for approximate membership queries, such as the Bloom Filter, further enhances the protocol’s scalability. While our current implementation employs efficient techniques, further advancements in data structures may provide additional improvements in scalability and performance without compromising the level of privacy protection.

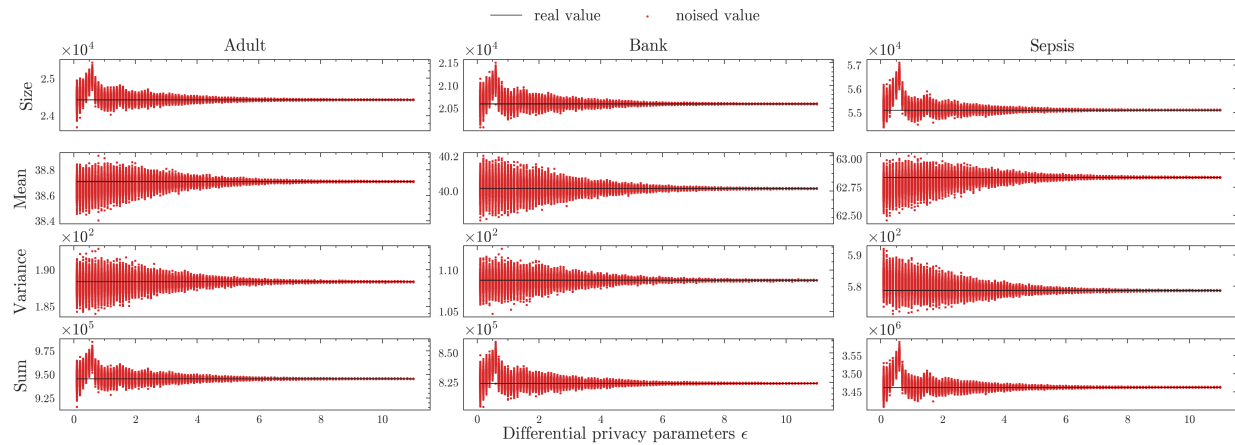


Fig. 8. Approximate intersection-based analysis of three public datasets (Adult, Bank, and Sepsis) with DP parameter (ϵ). Although these associated values follow different distributions, the analysis using intersection size, mean, variance, and sum confirms the adaptability and feasibility of our scheme.

B. Multiple Queries

A challenge in our protocol arises from the privacy budget consumption during multiple queries due to the sequential composition of DP. One possible solution is to fix the noised intersection output across multiple PSI tasks between the same parties on the same datasets, otherwise, the difference between the multiple queries will leak additional information, thereby reducing privacy protection. This allows the randomness of our protocol to be fixed after initialization, including the mapping function in the OPRF instance, the random seed in the random obfuscation, and the hash function in membership conflict. However, it may still require careful fine-tuning to balance privacy and data utility in multiple queries.

Additionally, the privacy budget is controlled only by $\mathcal{S}$, so it can be carefully adjusted in only one party according to analytical applications in another party.

C. One-Sided Output and Unbalanced Setting

Our protocol has a major shortcoming in that it only supports one-sided output in the ideal functionality of Algorithm 2, where in many settings both parties may need to obtain the output. Simply, $\mathcal{R}$ can directly deliver the result of the intersection-based analysis to another party if the protocol guarantees semi-honest security. Nevertheless, it is worth noting that $\mathcal{R}$ cannot directly send the noised intersection to the sender due to privacy within the intersection must be taken into consideration. In detail, a false positive the receiver obtains is definitely a true negative to the sender, and vice versa. Therefore, if $\mathcal{S}$ obtain Z' , he can infer the elements that belong to Z .

The proposed scheme naturally supports unbalanced dataset settings due to its reliance on fixed-key OPRF, which operates independently of the relative sizes of the input sets. Specifically, the parameters p_r and p_c are solely determined by the privacy budget ϵ , and are independent of the sizes of the datasets held by each party. Although highly unbalanced inputs may lead to a higher false positive rate and potentially degrade the accuracy of approximate results, this does not

affect the security guarantees or privacy protection ensured by our protocol.

D. Malicious Model Considerations

In the malicious setting, three aspects are noteworthy. The first concerns the OPRF, which only guarantees semi-honest security in our protocol. Malicious behaviors could be mitigated by employing verifiable or maliciously secure OPRF variants at additional cost. The second aspect involves a malicious $\mathcal{S}$ that may deviate from the prescribed obfuscation or AMQ construction. Such deviation primarily disrupts correctness but does not leak information about $\mathcal{R}$'s input, since no output is delivered to $\mathcal{S}$. Third, a malicious $\mathcal{R}$ may attempt to query arbitrary inputs and test them against the AMQ, effectively turning the protocol into a noisy membership oracle. However, such leakage is fundamentally limited by the DP guarantees of our mechanism, which bound $\mathcal{R}$'s advantage under the parameter ϵ .

IX. CONCLUSION

By combining OPRF with the RR mechanism, Approx-PSI presents an efficient and privacy-preserving solution under the semi-honest model that achieves significant improvements in computational and communication efficiency while enabling flexible analysis over the noised intersection.

Nevertheless, several limitations remain. First, the protocol outputs a differentially private (noised) intersection, which is unsuitable for applications requiring exact results. Additionally, the choice of privacy budget currently requires manual tuning without adapting to data characteristics or analysis goals. To expand the framework's capabilities, future work may explore advanced analysis functions beyond deterministic aggregates. Another direction is to extend the protocol to interactive or iterative settings, where cumulative privacy loss must be managed across multiple dependent queries through appropriate composition strategies. In addition, adaptive mechanisms for selecting the privacy budget ϵ based on task requirements or data characteristics can improve utility while maintaining formal privacy guarantees.

ACKNOWLEDGMENT

This work was supported in part by National Cryptologic Science Fund of China (No. 2025NCSF02058) and by National Natural Science Foundation of China (No. 62572437).

REFERENCES

- [1] G. Garimella, B. Pinkas, M. Rosulek, N. Trieu, and A. Yanai, "Oblivious key-value stores and amplification for private set intersection," in *Advances in Cryptology—CRYPTO 2021: 41st Annual International Cryptology Conference, CRYPTO 2021, Virtual Event, August 16–20, 2021, Proceedings, Part II* 41. Springer, 2021, pp. 395–425.
- [2] S. Raghuraman and P. Rindal, "Blazing Fast PSI from Improved OKVS and Subfield VOLE," in *Proceedings of the 2022 ACM SIGSAC Conference on Computer and Communications Security*. Association for Computing Machinery, 2022, pp. 2505–2517.
- [3] K. Cong, R. C. Moreno, M. B. da Gama, W. Dai, I. Iliashenko, K. Laine, and M. Rosenberg, "Labeled PSI from homomorphic encryption with reduced computation and communication," in *CCS '21: 2021 ACM SIGSAC Conference on Computer and Communications Security, Virtual Event, Republic of Korea, November 15 - 19, 2021*, Y. Kim, J. Kim, G. Vigna, and E. Shi, Eds. ACM, 2021, pp. 1135–1150.
- [4] B. Pinkas, T. Schneider, O. Tkachenko, and A. Yanai, "Efficient Circuit-Based PSI with Linear Communication," in *Advances in Cryptology – EUROCRYPT 2019*, Y. Ishai and V. Rijmen, Eds., vol. 11478. Cham: Springer International Publishing, 2019, pp. 122–153.
- [5] N. Chandran, D. Gupta, and A. Shah, "Circuit-psi with linear complexity via relaxed batch oprf," *Proceedings on Privacy Enhancing Technologies*, 2022.
- [6] N. Trieu, K. Shehata, P. Saxena, R. Shokri, and D. Song, "Epione: Lightweight contact tracing with strong privacy," *CoRR*, vol. abs/2004.13293, 2020. [Online]. Available: <https://arxiv.org/abs/2004.13293>
- [7] D. Kales, C. Rechberger, T. Schneider, M. Senker, and C. Weinert, "Mobile private contact discovery at scale," in *28th USENIX Security Symposium (USENIX Security 19)*, 2019, pp. 1447–1464.
- [8] L. Hetz, T. Schneider, and C. Weinert, "Scaling mobile private contact discovery to billions of users," in *Computer Security – ESORICS 2023*, G. Tsudik, M. Conti, K. Liang, and G. Smaragdakis, Eds. Cham: Springer Nature Switzerland, 2024, pp. 455–476.
- [9] Y. Huang, D. Evans, and J. Katz, "Private set intersection: Are garbled circuits better than custom protocols?" in *NDSS*, 2012.
- [10] B. Pinkas, T. Schneider, and M. Zohner, "Scalable private set intersection based on ot extension," *ACM Transactions on Privacy and Security (TOPS)*, vol. 21, no. 2, pp. 1–35, 2018.
- [11] E. D. Cristofaro, P. Gasti, and G. Tsudik, "Fast and private computation of cardinality of set intersection and union," in *Cryptology and Network Security, 11th International Conference, CANS 2012, Darmstadt, Germany, December 12–14, 2012. Proceedings*, J. Pieprzyk, A. Sadeghi, and M. Manulis, Eds., vol. 7712. Springer, 2012, pp. 218–231.
- [12] H. Chen, K. Laine, and P. Rindal, "Fast private set intersection from homomorphic encryption," in *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security, CCS 2017, Dallas, TX, USA, October 30 - November 03, 2017*, B. Thuraisingham, D. Evans, T. Malkin, and D. Xu, Eds. ACM, 2017, pp. 1243–1255.
- [13] B. Kacsmar, B. Khurram, N. Lukas, A. Norton, M. Shafieinejad, Z. Shang, Y. Baseri, M. Sepehri, S. Oya, and F. Kerschbaum, "Differentially private two-party set operations," in *IEEE European Symposium on Security and Privacy, EuroS&P 2020, Genoa, Italy, September 7–11, 2020*. IEEE, 2020, pp. 390–404.
- [14] Y. He, X. Tan, J. Ni, L. T. Yang, and X. Deng, "Differentially private set intersection for asymmetrical ID alignment," *IEEE Trans. Inf. Forensics Secur.*, vol. 17, pp. 3479–3494, 2022.
- [15] S. Dittmer, Y. Ishai, S. Lu, R. Ostrovsky, M. Elsabahg, N. Kiourtis, B. Schulte, and A. Stavrou, "Function secret sharing for PSI-CA: with applications to private contact tracing," *IACR Cryptol. ePrint Arch.*, p. 1599, 2020.
- [16] M. Wu and T. H. Yuen, "Efficient unbalanced private set intersection cardinality and user-friendly privacy-preserving contact tracing," in *32nd USENIX Security Symposium (USENIX Security 23)*. Anaheim, CA: USENIX Association, Aug. 2023, pp. 283–300.
- [17] M. Ion, B. Kreuter, E. Nergiz, S. Patel, S. Saxena, K. Seth, D. Shanahan, and M. Yung, "Private intersection-sum protocol with applications to attributing aggregate ad conversions," *Cryptology ePrint Archive*, 2017.
- [18] M. Ion, B. Kreuter, A. E. Nergiz, S. Patel, S. Saxena, K. Seth, M. Raykova, D. Shanahan, and M. Yung, "On Deploying Secure Computing: Private Intersection-Sum-with-Cardinality," in *2020 IEEE European Symposium on Security and Privacy (EuroS&P)*, 2020, pp. 370–389.
- [19] P. Miao, S. Patel, M. Raykova, K. Seth, and M. Yung, "Two-sided malicious security for private intersection-sum with cardinality," in *Advances in Cryptology - CRYPTO 2020 - 40th Annual International Cryptology Conference, CRYPTO 2020, Santa Barbara, CA, USA, August 17–21, 2020, Proceedings, Part III*, ser. Lecture Notes in Computer Science, D. Micciancio and T. Ristenpart, Eds., vol. 12172. Springer, 2020, pp. 3–33.
- [20] J. H. M. Ying, S. Cao, G. S. Poh, J. Xu, and H. W. Lim, "Psi-Stats: Private Set Intersection Protocols Supporting Secure Statistical Functions," in *Applied Cryptography and Network Security*. Springer International Publishing, 2022, pp. 585–604.
- [21] G. Garimella, P. Mohassel, M. Rosulek, S. Sadeghian, and J. Singh, "Private set operations from oblivious switching," in *Public-Key Cryptography - PKC 2021 - 24th IACR International Conference on Practice and Theory of Public Key Cryptography, Virtual Event, May 10–13, 2021, Proceedings, Part II*, ser. Lecture Notes in Computer Science, J. A. Garay, Ed., vol. 12711. Springer, 2021, pp. 591–617.
- [22] B. Pinkas, M. Rosulek, N. Trieu, and A. Yanai, "PSI from paxos: Fast, malicious private set intersection," in *Advances in Cryptology - EUROCRYPT 2020 - 39th Annual International Conference on the Theory and Applications of Cryptographic Techniques, Zagreb, Croatia, May 10–14, 2020, Proceedings, Part II*, ser. Lecture Notes in Computer Science, A. Canteaut and Y. Ishai, Eds., vol. 12106. Springer, 2020, pp. 739–767.
- [23] X. Guo, Y. Han, Z. Liu, D. Wang, Y. Jia, and J. Li, "Birds of a feather flock together: How set bias helps to deanonymize you via revealed intersection sizes," in *31st USENIX Security Symposium (USENIX Security 22)*, 2022, pp. 1487–1504.
- [24] S. Lu, H. Dong, Z. Li, and L. T. Yang, "Not just summing: The identifier leakage of private-join-and-compute and its improvement," *IEEE Transactions on Dependable and Secure Computing*, pp. 1–13, 2024.
- [25] B. Jiang, J. Du, and Q. Yan, "Anonpsi: An anonymity assessment framework for PSI," in *31st Annual Network and Distributed System Security Symposium, NDSS 2024, San Diego, California, USA, February 26 - March 1, 2024*. The Internet Society, 2024.
- [26] D. Dachman-Soled, T. Malkin, M. Raykova, and M. Yung, "Efficient robust private set intersection," *International Journal of Applied Cryptography*, vol. 2, no. 4, pp. 289–303, 2012.
- [27] Z. Liang, W. Liu, H. Feng, F. Han, L. Peng, L. Peng, C. Li, G. Xu, L. Zhang, and F. Zhang, "PsiBench: Pragmatic benchmark of two-party private set intersection." *Cryptology ePrint Archive*, Paper 2020/1541, 2020. [Online]. Available: <https://eprint.iacr.org/2020/1541>
- [28] Á. Kiss, J. Liu, T. Schneider, N. Asokan, and B. Pinkas, "Private set intersection for unequal set sizes with mobile applications," in *Privacy Enhancing Technologies Symposium*. De Gruyter, 2017, pp. 177–197.
- [29] A. C. D. Resende and D. F. Aranha, "Faster unbalanced private set intersection," in *Financial Cryptography and Data Security: 22nd International Conference, FC 2018, Nieuwpoort, Curaçao, February 26–March 2, 2018, Revised Selected Papers 22*. Springer, 2018, pp. 203–221.
- [30] R. A. Mahdavi, N. Lukas, F. Ebrahimiaghazani, T. Humphries, B. Kacsmar, J. Premkumar, X. Li, S. Oya, E. Amjadian, and F. Kerschbaum, "[PEPSI]: Practically efficient private set intersection in the unbalanced setting," in *33rd USENIX Security Symposium (USENIX Security 24)*, 2024, pp. 6453–6470.
- [31] M. Purcell, Y. Li, and K. S. Ng, "Split, count, and share: a differentially private set intersection cardinality estimation protocol," in *Proceedings of the Thirty-Ninth Conference on Uncertainty in Artificial Intelligence*, ser. Proceedings of Machine Learning Research, R. J. Evans and I. Shpitser, Eds., vol. 216. PMLR, 31 Jul–04 Aug 2023, pp. 1684–1694.
- [32] X. Peng, Y. Wang, W. Liu, L. Peng, F. Han, Z. Gu, J. Sun, and Y. Hong, "Efficient differentially private set intersection," *Cryptology ePrint Archive*, Paper 2024/1239, 2024. [Online]. Available: <https://eprint.iacr.org/2024/1239>
- [33] J. Du, T. Ji, J. Cui, L. Zhang, Y. Lu, and P. Duan, "Dp-psi: Private and secure set intersection," *arXiv preprint arXiv:2208.13249*, 2022.
- [34] C. Dwork, "Differential privacy," in *Automata, languages and programming, 33rd international colloquium, ICALP 2006, venice, italy, july 10–14, 2006, proceedings, part II*, vol. 4052. Springer, 2006, pp. 1–12.

- [35] C. Dwork and J. Lei, "Differential privacy and robust statistics," in *Proceedings of the forty-first annual ACM symposium on Theory of computing*, 2009, pp. 371–380.
- [36] S. L. Warner, "Randomized Response: A Survey Technique for Eliminating Evasive Answer Bias," *Journal of the American Statistical Association*, vol. 60, no. 309, pp. 63–69, 1965.
- [37] B. Becker and R. Kohavi, "Adult," UCI Machine Learning Repository, 1996, DOI: <https://doi.org/10.24432/C5XW20>.
- [38] S. Moro, P. Rita, and P. Cortez, "Bank Marketing," UCI Machine Learning Repository, 2014, DOI: <https://doi.org/10.24432/C5K306>.
- [39] D. Chicco and G. Jurman, "Sepsis Survival Minimal Clinical Records," UCI Machine Learning Repository, 2023, DOI: <https://doi.org/10.24432/C53C8N>.



Zhan Qin (Member, IEEE) received the PhD degree from the Department of Computer Science and Engineering, University at Buffalo, State University of New York, in 2017. He is currently an Assistant Professor with the School of Cyber Science and Technology, Zhejiang University. His research interests include intersection of AI security and differential privacy.



Jian Long is currently pursuing his Ph.D. in software engineering at the College of Computer Science and Technology, Zhejiang University, Hangzhou, China. His research interests include secure computing and privacy-preserving data sharing.



Ziyuan Liang received his Ph.D. degree in information science and electronic engineering at the School of Cyberspace Research, Zhejiang University, Hangzhou, China in 2023. His research interests include secure computing, Intel SGX technique, and side-channel attacks.



Chao Li is currently pursuing the Master degree in software engineering at the School of Cyber Science and Technology, Zhejiang University. His research interests include multiparty computation and homomorphic encryption.



Zhang Fan (Member, IEEE) received the Ph.D. degree from the Department of Computer Science and Engineering, University of Connecticut, CT, USA, in 2011. He is currently a Full Professor with the College of Computer Science and Technology, Zhejiang University, Hangzhou, China. His research interests include system security, hardware security, network security, cryptography, and computer architecture.